

Richard Pu

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EDUCATION

University of Waterloo

Waterloo, ON

Candidate for B.A.Sc. in Computer Engineering (Honours, Co-op)

Sept. 2026 – June 2031

- **Stream 4 Co-op:** Seeking first 4-month internship (**Jan. 2027 – Apr. 2027**)
- **Relevant Coursework:** Fundamentals of Programming (C++), Linear Circuits, Digital Circuits, Linear Algebra

Thornhill Secondary School

Thornhill, ON

Ontario Secondary School Diploma

Sept. 2022 – June 2026

- 95% Cumulative Average (4.0 GPA) – Academic Honours Across All Disciplines

TECHNICAL PROJECTS

Smart Chess Board | NVIDIA Jetson Orin Nano, Python, OpenCV, Flask

2025

- Rebuilt an abandoned school chessboard project on a Jetson Orin Nano after a prior Raspberry Pi/Arduino design failed, wiring 10 buttons, 64 addressable LEDs, and an OLED display directly to GPIO.
- Integrated Stockfish (offline AI) and the Lichess API (online play) with a real-time Flask dashboard for move history and custom LED themes; cut startup time by over 50%.
- Redesigning and 3D-printed the enclosure for airflow, component access, and long-term repairability.

Real-Time CV “Red Light, Green Light” | Python, OpenCV, YOLOv8, TensorRT, Jetson

2026

- Built a self-officiating, multi-threaded game on a Jetson Orin Nano that tracks every player via YOLOv8-pose accelerated with TensorRT.
- Designed an architecture separating camera capture, pose inference, and game logic so display frame rate stayed decoupled from inference latency.
- Drove a servo-controlled head and camera over I2C/GPIO, with per-player elimination detection by clothes and real-time audio TTS, running in a Docker container for reproducible deployment on the Jetson.

BenumZombs – 2D Survival/Tower-Defense Game | Java, Swing/AWT

2026

- Designed a full Java game architecture (loop, rendering, collision, resource, and building systems) supporting a day/night cycle and four upgradeable tower types.
- Implemented polymorphic entity and tower classes using object-oriented design patterns, managing dynamic state transitions, resource economies, and automated targeting systems.
- Engineered a scalable wave-spawning algorithm and grid-based collision detection system in Java/Swing, maintaining responsive rendering and stable performance under high entity counts.

EXPERIENCE

Lifeguard & Swim Instructor

Vaughan, ON

City of Vaughan Aquatics

Sept. 2024 – Jun. 2026

- Enforced pool safety regulations in a safety-critical environment, responding to first-aid incidents and de-escalating conflicts with the public.
- Delivered one-on-one and group swim lessons across age groups, completing 80+ hours of National Lifeguard/Swim Instructor certification (water rescue, First Aid/CPR-C).

LEADERSHIP & ACTIVITIES

Co-President, Tech Crew

Thornhill, ON

Thornhill Secondary School

Grades 9 – 12

- Led a student-run A/V team operating audio, lighting, and equipment for 30+ school events; grew active membership from 5 to 20 and built a shared calendar and inventory system that made the team self-sustaining.

Co-President, Announcements Team

Thornhill, ON

Thornhill Secondary School

Grades 9 – 12

- Identified a school-wide information gap and launched a digital Instagram announcements platform, reaching 50,000+ views in 90 days.

TECHNICAL SKILLS

Languages: Python, C++, Java, TypeScript, HTML/CSS

Frameworks & Libraries: React, Flask, OpenCV, YOLOv8, TensorRT

Hardware & Tools: NVIDIA Jetson Orin Nano, Arduino, Raspberry Pi, Docker, Git, Linux